

SAMA SUMANTH REDDY

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SUMMARY

Machine Learning Engineer with hands-on experience building end-to-end ML and computer vision solutions, including data preprocessing, feature engineering, neural network model training (CNN-based object detection with YOLOv8), hyperparameter optimization, model evaluation, and deployment. Skilled in Python, SQL, Scikit-learn, and OpenCV, with practical exposure to ensemble learning, model optimization workflows, and performance evaluation (Precision/Recall/F1, ROC-AUC, cross-validation). Strong foundation in neural network architectures and a fast learner eager to build expertise in AI/ML compiler toolchains and large-scale model optimization for next-generation compute platforms.

SKILLS

Languages / Querying: Python, SQL

Machine Learning / Deep Learning: ANNs, CNNs, YOLOv8 (PyTorch-based object detection), Computer Vision, Image Classification, OpenCV, Linear/Logistic Regression, Decision Trees, Random Forest, Naive Bayes, Gradient Boosting, XGBoost, AdaBoost, SVM, KNN, K-Means, DBSCAN, Hierarchical Clustering, GMM, PCA, t-SNE

Data Wrangling / Cleaning: Missing Value Handling, Outlier Removal, Class-Imbalance Correction, Feature Engineering, Label/One-Hot Encoding

Model Optimization & Evaluation: Hyperparameter Tuning, GridSearchCV, Cross-Validation, ROC-AUC, Precision/Recall/F1, Confusion Matrix, Bias-Variance Tradeoff

Model Deployment: Joblib Pipeline Deployment for Real-Time Inference

Reinforcement Learning: Q-Learning, SARSA, Markov Decision Processes, Bellman Equation

Tools / Frameworks: Pandas, NumPy, Scikit-learn, MySQL, Git, GitHub, Jupyter Notebook

PROJECTS

Debris-Based Disaster Victim Identification Using Deep Learning

Python, YOLOv8 (CNN-based object detection), OpenCV, Scikit-learn, Decision Tree, Random Forest, SVM, Matplotlib, Seaborn

- Preprocessed and augmented multi-type disaster-scenario image datasets (resizing, normalization, augmentation) using OpenCV to support neural network training for human victim detection.
- Trained a YOLOv8 convolutional neural network for real-time object detection and victim localization in occluded debris environments, achieving reliable bounding-box precision.
- Optimized model performance by integrating ensemble classifiers (Decision Tree, Random Forest, SVM) as a post-detection decision layer, improving F1-score by 12%.
- Performed systematic performance evaluation using Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix; applied GridSearchCV for hyperparameter optimization on class-imbalanced outputs.

AI-Driven Health Diagnosis System for Symptom-Based Disease Prediction

Python, Pandas, NumPy, Scikit-learn, Joblib, MySQL, Jupyter Notebook

- Cleaned and preprocessed a 1,000+ record healthcare symptom dataset, handling missing values, outlier removal, and class-imbalance correction.
- Engineered categorical features via label and one-hot encoding; performed feature selection to identify top predictive symptom indicators.
- Built, benchmarked, and optimized five model architectures (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Naive Bayes), achieving up to 0.93 accuracy via 5-fold cross-validation.
- Deployed the best-performing model as a Joblib pipeline for real-time inference and disease prediction.

EDUCATION

Vellore Institute of Technology, Vellore

M.Tech Integrated Software Engineering, CGPA: 8.09

Aug 2021 – May 2026

Alphores Junior College, Karimnagar

Telangana State Board of Intermediate Education (MPC), Score: 972

June 2019 – April 2021

CERTIFICATIONS

- Career Essentials in Data Analysis — Microsoft & LinkedIn
- Exploratory Data Analysis — Infosys